

Computer Networks

CM3108

UNIT 4 : Network Topologies And Controlling Devices

4.1 : Network Topologies

- **Priti Tamhane.**

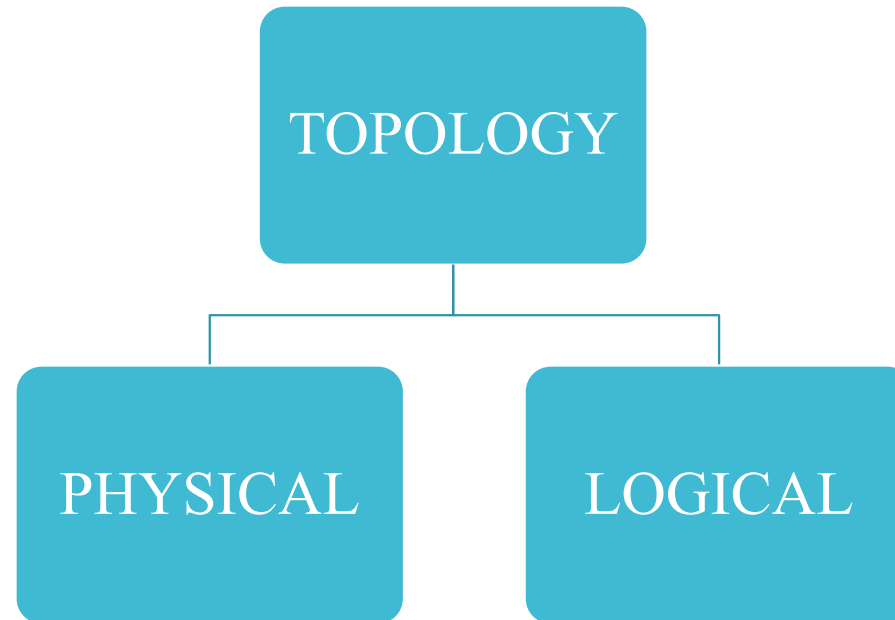
Student at Department of Computer

Engineering, Government Polytechnic, Pune.

Under the guidance of **A. A. SHAIKH Mam.**

NETWORK TOPOLOGY

Network topology is basically the arrangement of nodes in a computer network.



PHYSICAL TOPOLOGY



The Physical topology refers to how the nodes of network are physically connected, i.e., the layout of cabling, the locations of nodes, and interconnection between the nodes and the cabling.

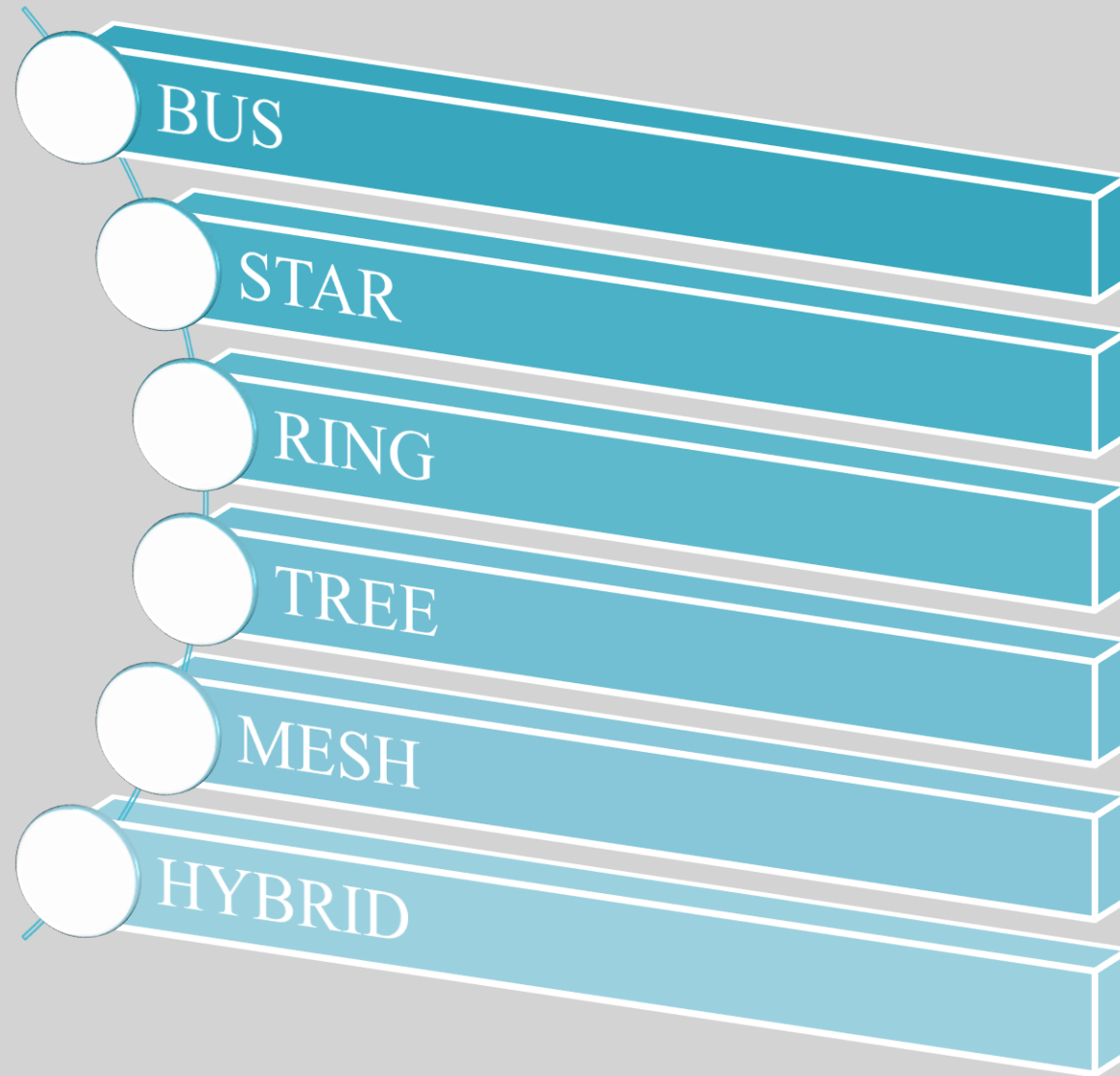
LOGICAL TOPOLOGY



The Logical topology refer to how the data is transmitted between the nodes, i.e., the way that the signals act on the network media, or the way that data passes through the network from one device to the next irrespective of physical interconnection of the devices.

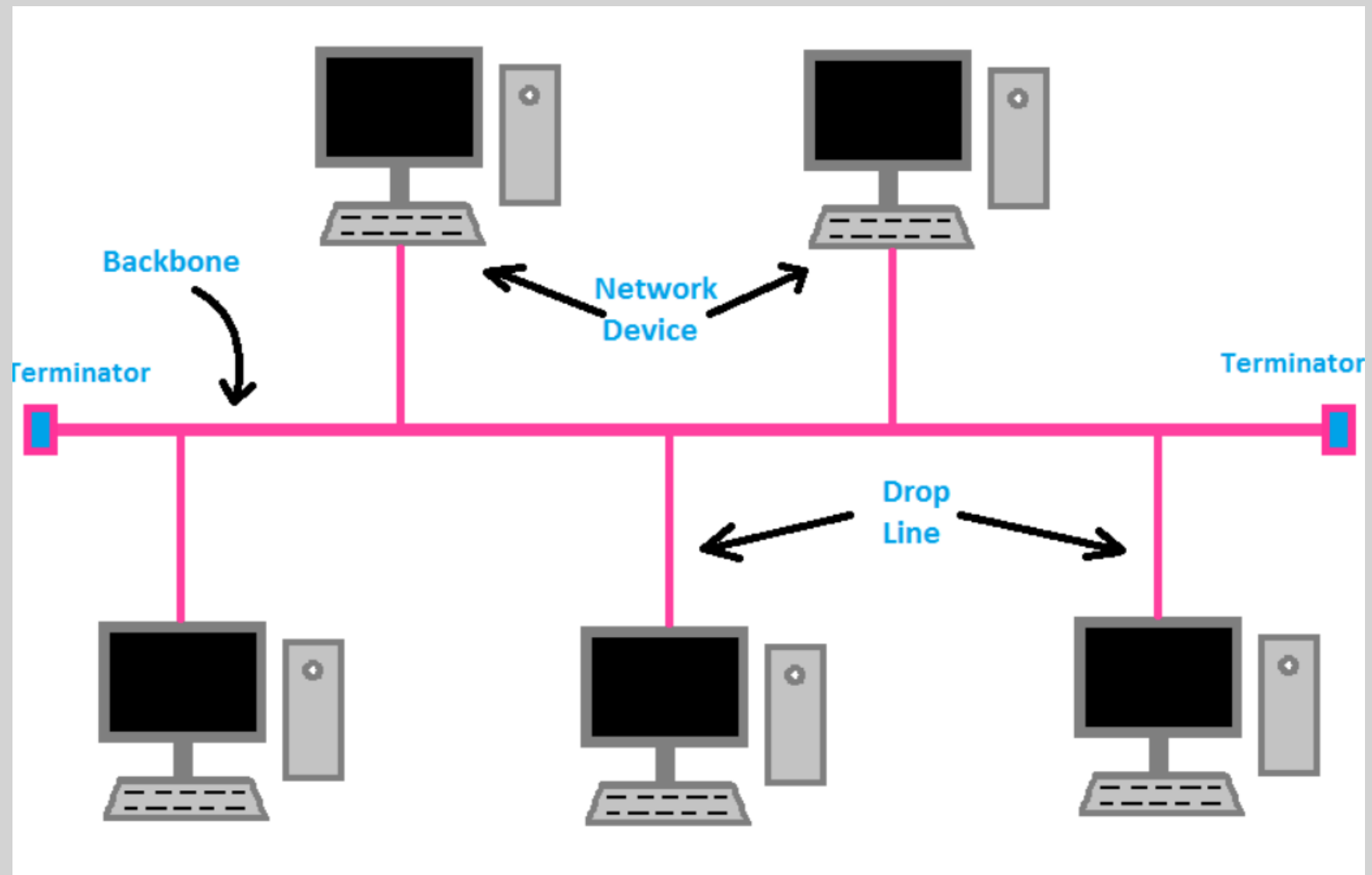
- A network's logical topology is not necessarily as same as the physical topology.
- In order to have an efficient system, the logical topology should be chosen

NETWORK TOPOLOGIES



BUS TOPOLOGY

- A bus topology is a type of network in which each node present for communication is connected to a wire. This topology is also known as the Line topology.
- This wire is responsible for the communication between all the nodes in the network, and it is termed as 'backbone'.
- All the devices are connected to the backbone by drop lines and taps.
- Drop line is the connection between devices and the cable.
- Tap is the splitter that cut the main link.



DATA TRANSMISSION IN BUS TOPOLOGY

- Data transmission in bus topology is bidirectional.
- When a device sends the data, all the devices will receive that data. That means the data is broadcast in the bus topology.
- All devices on the cable constantly monitor for the data meant to them.
- When any device detects the data sent for it, it accepts the data from the cable and other devices will reject it. The destination address is sent along with the data.
- Terminators at the end of cable prevent the signals from reflecting back to the sender.
- Commonly, coaxial cable is used as transmission media in this topology.

ADVANTAGES

- Less expensive as only one cable is used for the transmission of the data.
- Installation is easy.
- Easy to reconfigure.
- Failure of any node will not affect to the others.

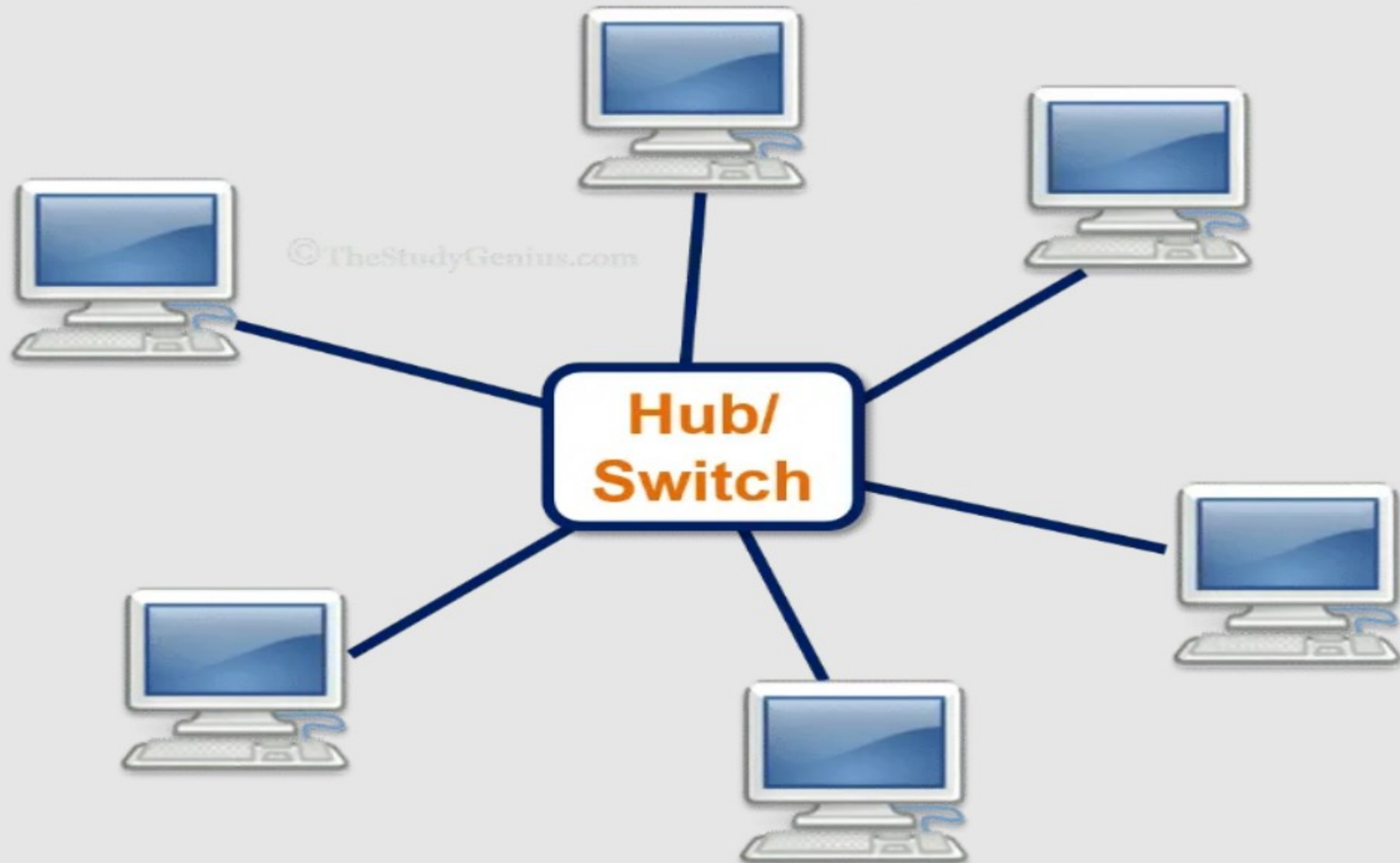
DISADVANTAGES

- Less secure as the data is broadcast.
- More the number of computers, less will be the strength of signals.
- Not fault tolerant.
- There are chances of collisions.
- If cable fails, entire network will collapse.

STAR TOPOLOGY

- Star topology is a type of network in which the centralized device is used.
- All the computers in the network are connected to a common point, i.e., the centralized device.
- In this topology, data transmission is done through this centralised device.
- This centralized device can be either hub or switch.

STAR Topology



DATA TRANSMISSION IN STAR TOPOLGY

- In star topology, when any device sends the data, it will first be received by the centralized device.
- Then this centralized device will transmit the data to the device whose address is given as the destination address.
- When the centralized device is hub, data sent by source machine will be sent to all the other devices connected in the network.
- On the other hand, when centralized device is switch, data sent by source machine will be sent to the destination device only.

ADVANTAGES

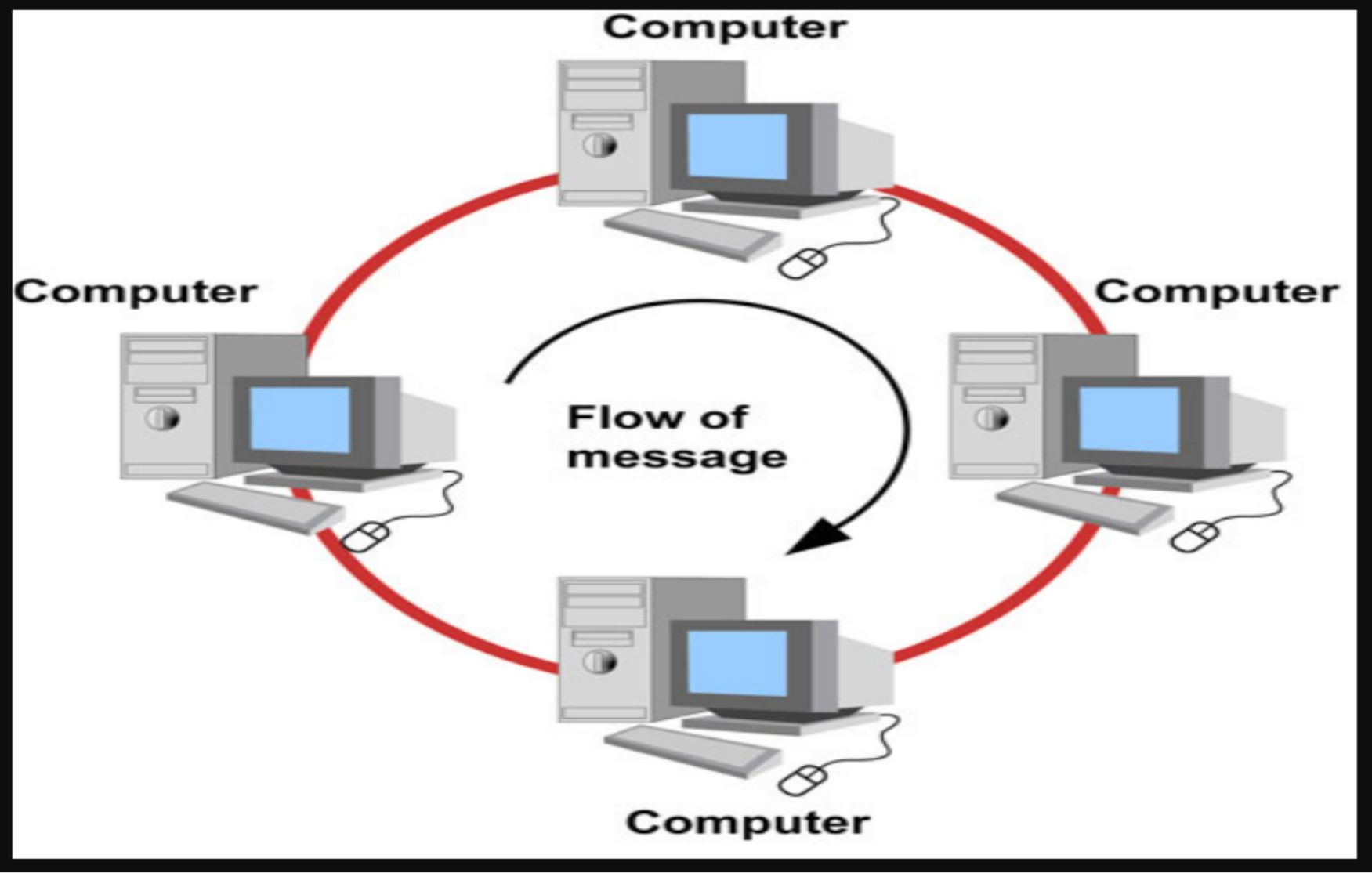
- Data is more secure when centralized device is switch.
- Fault tolerant and scalable.
- Easy to install and reconfigure.
- Many communications are possible at a time.

DISADVANTAGES

- If centralized device fails, entire network will fail.

RING TOPOLOGY

- Ring topology is a network similar to the bus topology, but in a closed loop.
- In this topology, every node is connected to the adjacent nodes.
- These nodes are peer-to-peer, that means each node has the equal rights.
- This topology is also known as the token ring topology, as the data is transmitted with help of the tokens.
- In ring topology, commonly, twisted pair cables and fibre optic cables are used for the transmission of the data.



DATA TRANSMISSION IN RING TOPOLOGY

- Data is transmitted only in one direction.
- When data is sent by any device, it is passed to every device until it reaches to its destination.
- When one device received the data instead of intended another device, its repeater then regenerates the data and passes them along.

ADVANTAGES

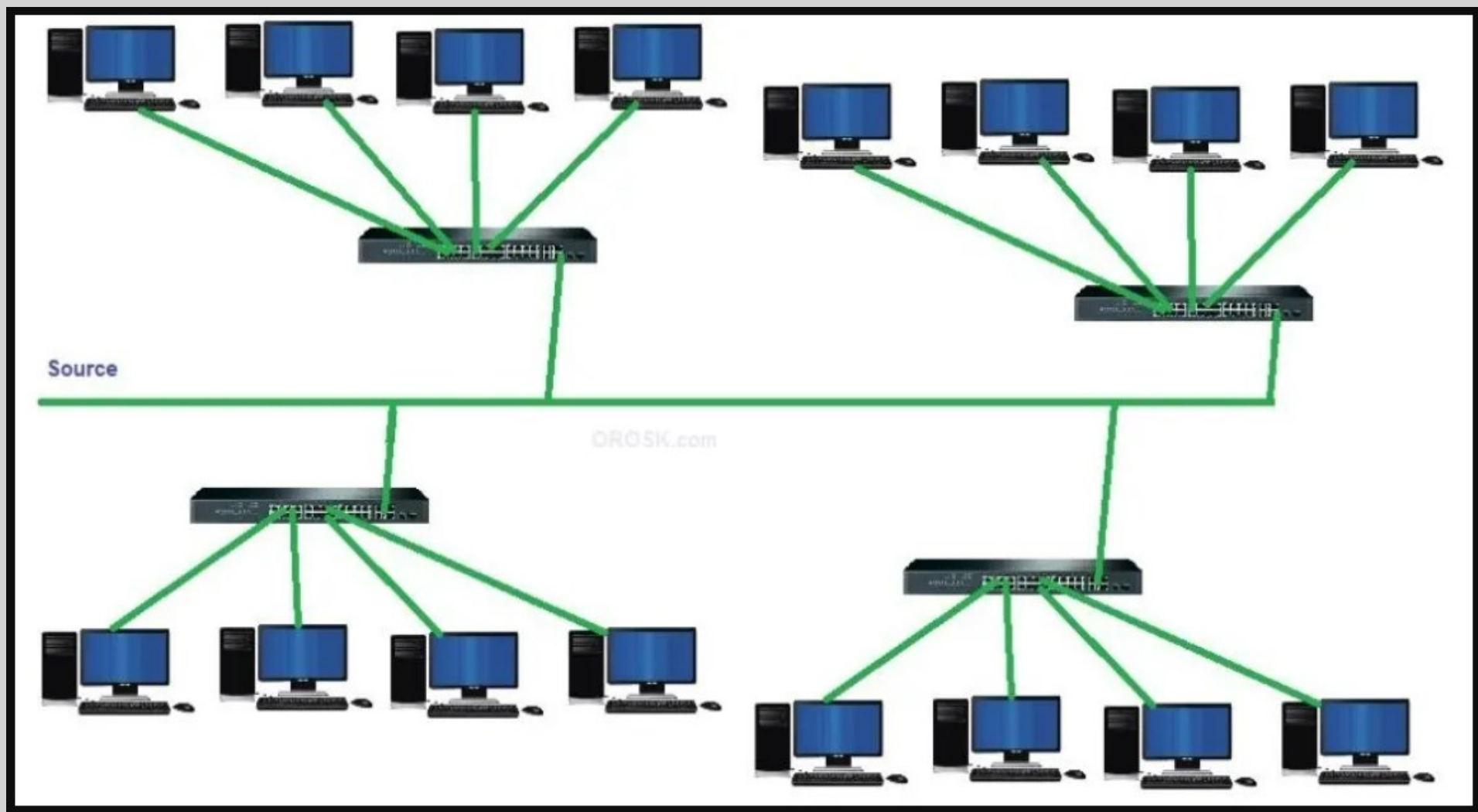
- No chances of collisions, as the data transmission is unidirectional.
- Easy to fault identification.
- Easy to install and reconfigure.
- Better than bus topology.

DISADVANTAGES

- Unidirectional traffic.
- If any of the system fails, entire network will fail.
- Less secure.

TREE TOPOLOGY

- A Tree topology is a type of network where many connected elements are arranged like the branches of a tree.
- Tree topology is also known as a star bus topology.
- In a tree topology, any two linked nodes can only have one mutual connection, hence there can only be one link between them.
- In this topology each node is related to the others in a hierarchy. Sometimes it is also called hierarchical topology, In this topological hierarchy, there are at least three distinct levels.
- Tree topologies are commonly used to arrange data in databases and workstations in corporate networks.



ADVANTAGES

- The data transfer rate is very high.
- Fault detection is easy
- Due to the utilization of Intelligent Hub/Switch, Tree Topology gives much greater performance than Bus Network.
- Scalable.
- If one or more nodes have failed in the network, they do not affect the entire network. And, the network can still continue to operate.
- Even if less space is available, new device can be added to the hub or switch. More space can be created through Star Network.

DISADVANTAGES

- Due to the large number of nodes in this network topology, its performance is greatly affected, as a result of which it becomes very slow.
- When the connector and hub are damaged, all the devices associated with it stop working.
- It is difficult to manage.
- Expensive to implement.

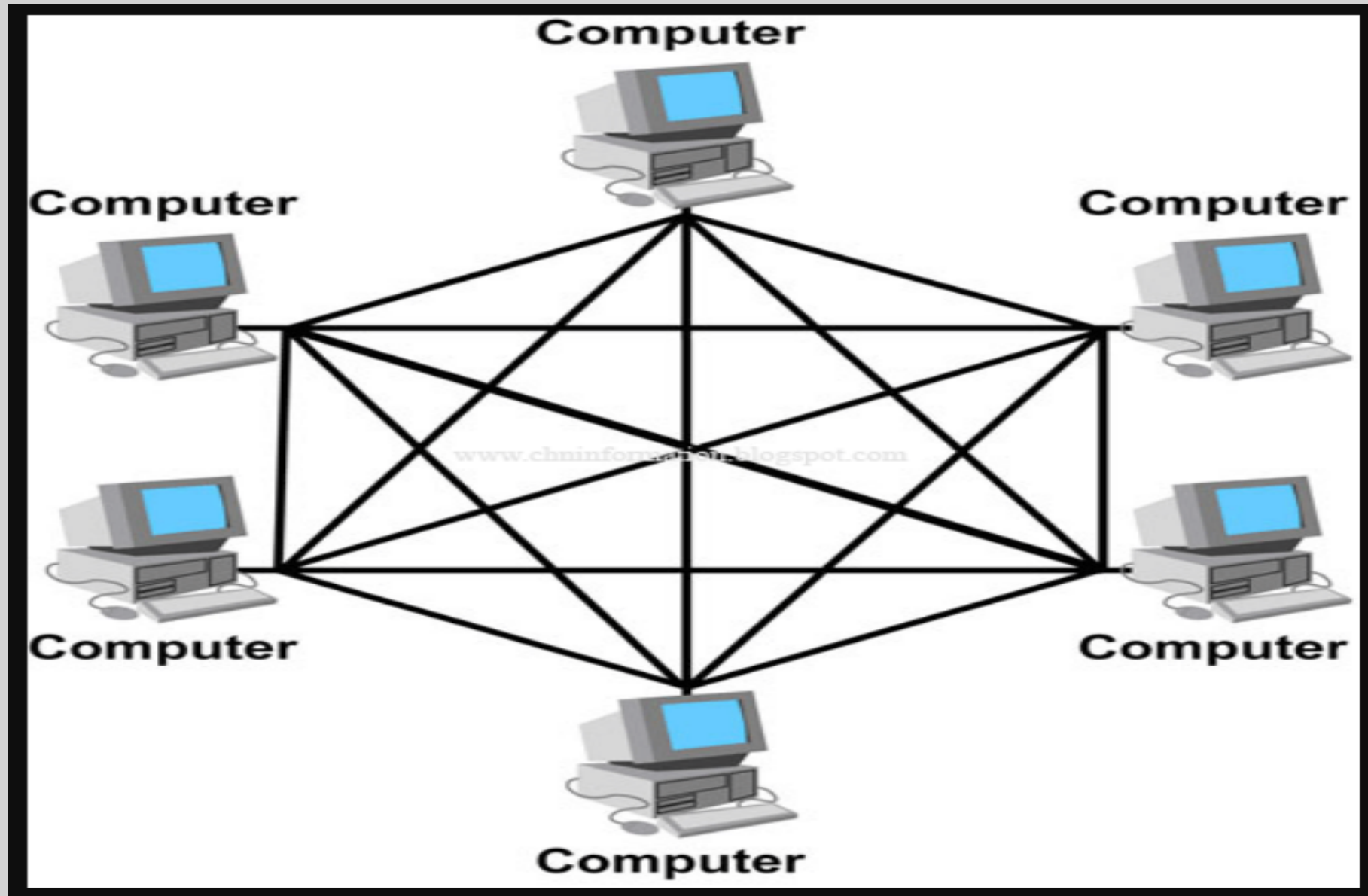
MESH TOPOLOGY

- Mesh topology is a network in which every system is connected to all the other systems.
- This topology is commonly used for wireless networks.

If there are 'n' nodes in the network.

then, each node will be connected to the 'n-1' nodes.

Total links will be $[n(n-1)/2]$.



ADVANTAGES

- Network traffic is avoided.
- Fault identification is easy.
- It is fault tolerant.
- It is robust.
- There is more security and privacy, as the message travels along a dedicated path.

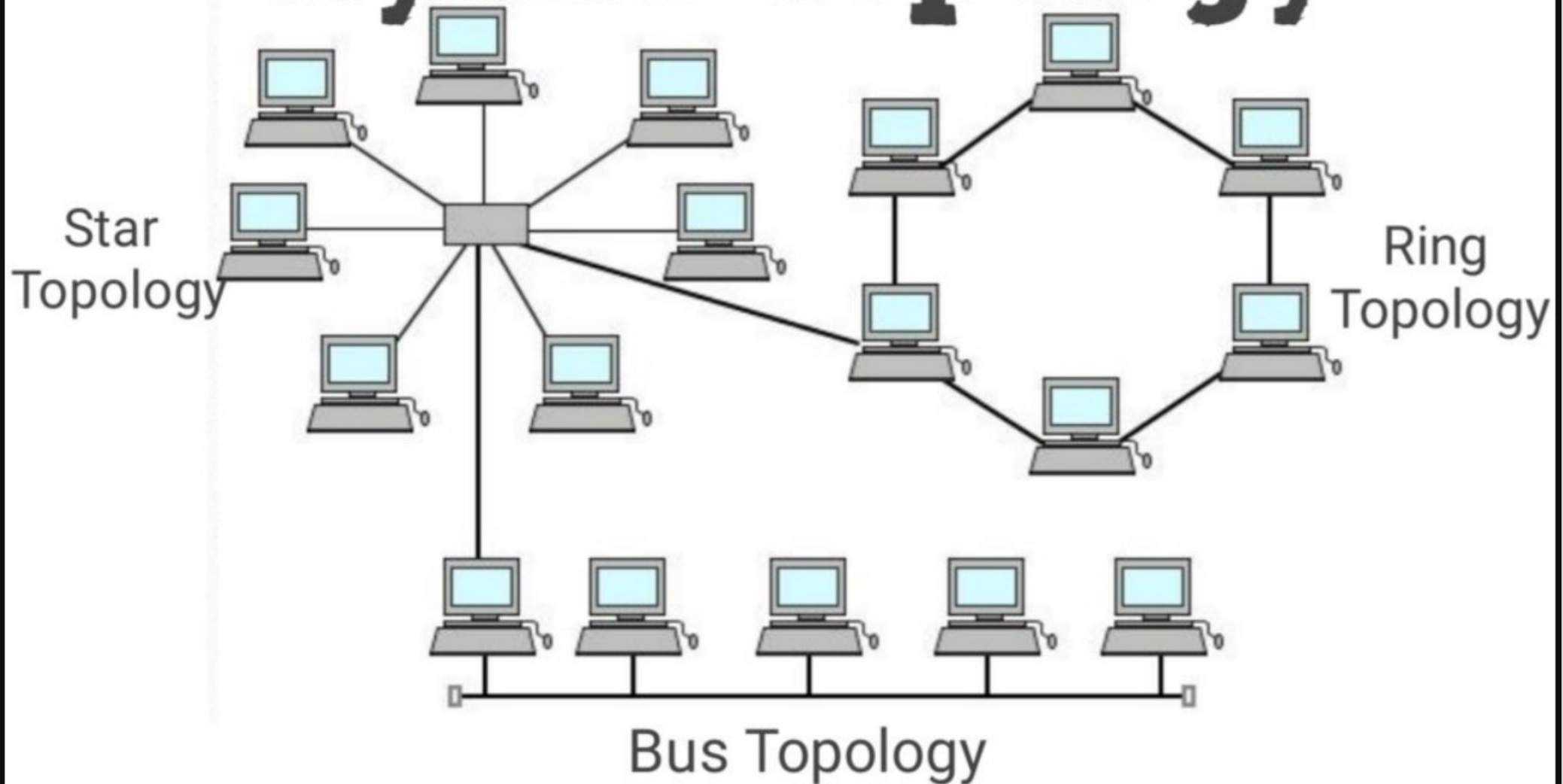
DISADVANTAGES

- Expensive
- Difficult to install and reconfigure.
- The amount of cabling and number of I/O ports required are very large. Since every device is connected to each device through dedicated links.

HYBRID TOPOLOGY

- A hybrid topology is a kind of network topology that is a combination of two or more network topologies, such as mesh topology, bus topology, and ring topology.
- Its usage and choice are dependent on its deployments and requirements like the performance of the desired network, and the number of computers, their location.

Hybrid Topology



ADVANTAGES

- It is more reliable as it has better fault tolerance.
- It has the ability to transfer data easily in between the different kinds of networks.
- It is scalable network.
- This topology can handle a large amount of data traffic.
- If any computer in the network fails to function then other computers in the network will not be affected.
- Hybrid topology gives the flexibility to join various types of independent networks.

DISADVANTAGES

- As the different topologies are linked in the hybrid topology, it is difficult to manage.
- Difficult to install and reconfigure.
- There are various types of devices are involved, so sometimes, it is expensive to create an intentional hybrid topology network.